



Optimization of FIR Filters Using Distributed Arithmetic for Low-Power MEMS Microphone Sensing Applications

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Introduction

Context: Next Level Research Green Aviation leverages Micro-Electro-Mechanical System (MEMS) microphones to study airflow transitions across bio-inspired airfoil surface geometries at the Aeronautical Wind Tunnel. The initiative aims to provide recommendations for zero-emission aircraft lifting surfaces.

Problem: Raw acoustic data contained significant background noise from facility operations, obscuring airflow transition characteristics.

Proposed Solution: A custom low-power digital bandpass Finite Impulse Response (FIR) filter was designed in Verilog to isolate target frequencies (10-13 kHz).

Optimization Goal: Replace standard Multiply-Accumulate (MAC) units with Distributed Arithmetic (DA), consisting of a Look-Up Table (LUT) architecture, to reduce power consumption and resource usage as well as align with sustainable aircraft design goals.

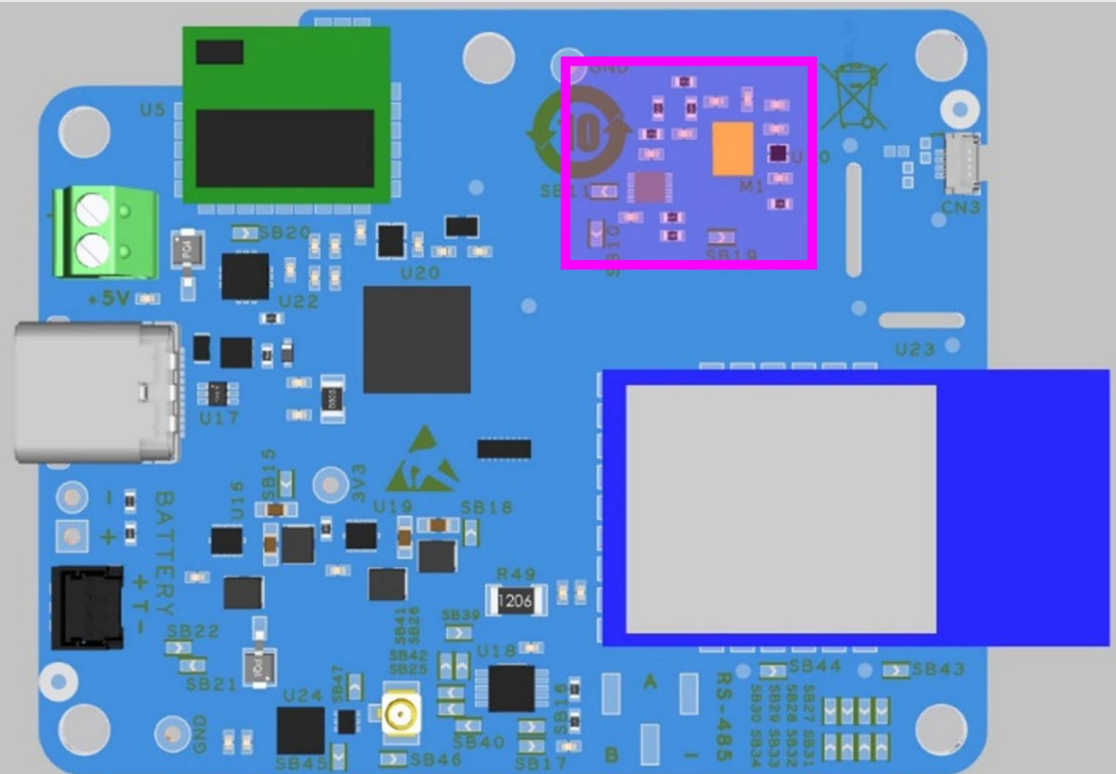


Figure 1. STEVAL-STWINBX1 Sensor Board With IMP23ABSU Analog MEMS Microphone.

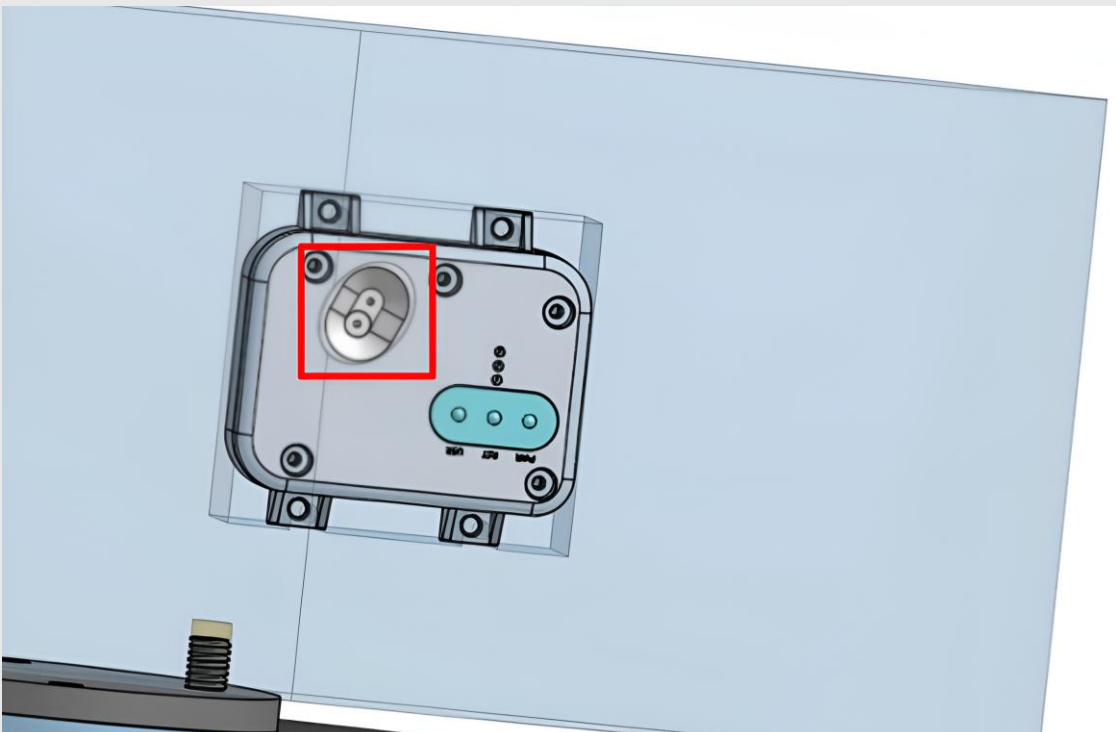


Figure 2. CAD of Airfoil for STEVAL-STWINBX1 With Venting Holes to Allow Airflow to Enter.

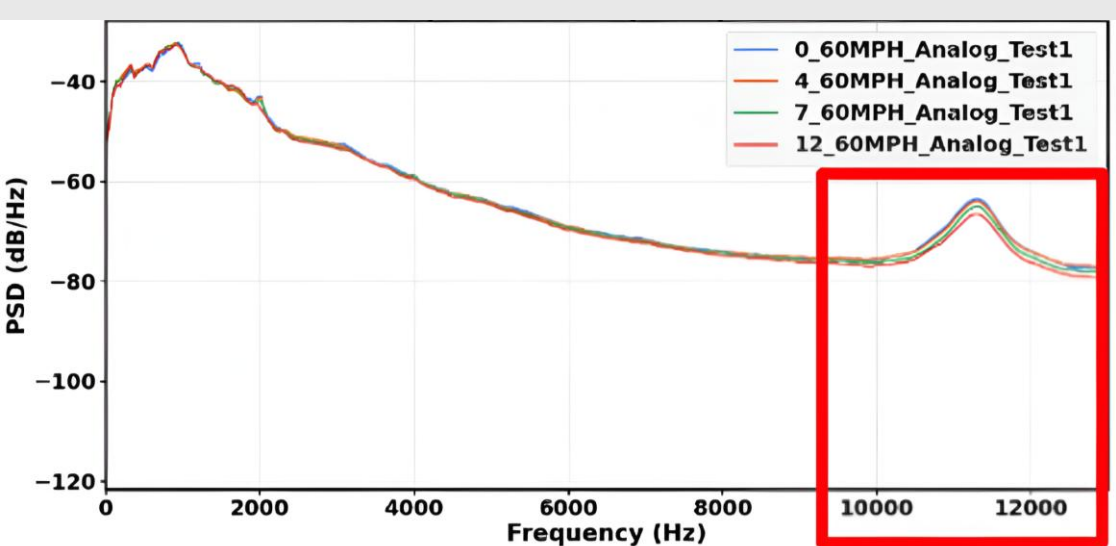


Figure 3. Power Spectral Density of Acoustic Data From Analog MEMS Microphone.

Methods

Data Collection & Requirements

- Captured acoustic airflow data during testing operations at the Aeronautical Wind Tunnel.
- Processed analog MEMS microphone recording files to establish baseline hardware and filter specifications.

Filter Modeling & Verification

- Designed a bandpass filter model in MATLAB Filter Designer to simulate the separation of target airflow transition frequencies from facility noise.
- Validated signal integrity in MATLAB using both synthetic test vectors and raw wind tunnel data.

Design Optimization

- Developed a logic flowchart and hardware architecture to map DA units to an FIR filter.
- Synthesized components in Quartus Prime to replace MAC blocks with LUTs and an Adder Tree.
- Validated system using Verilog testbenches executed in ModelSim.

Performance Evaluation

- Exported hardware outputs to MATLAB to compute Mean Square Error, Signal-to-Error Ratio, and Power Spectral Density.
- Used Quartus Power Analyzer and Fitter tools to benchmark hardware area and thermal power trade-offs in between the MAC-based FIR filter and DA-based FIR filter.

Standard FIR Filter

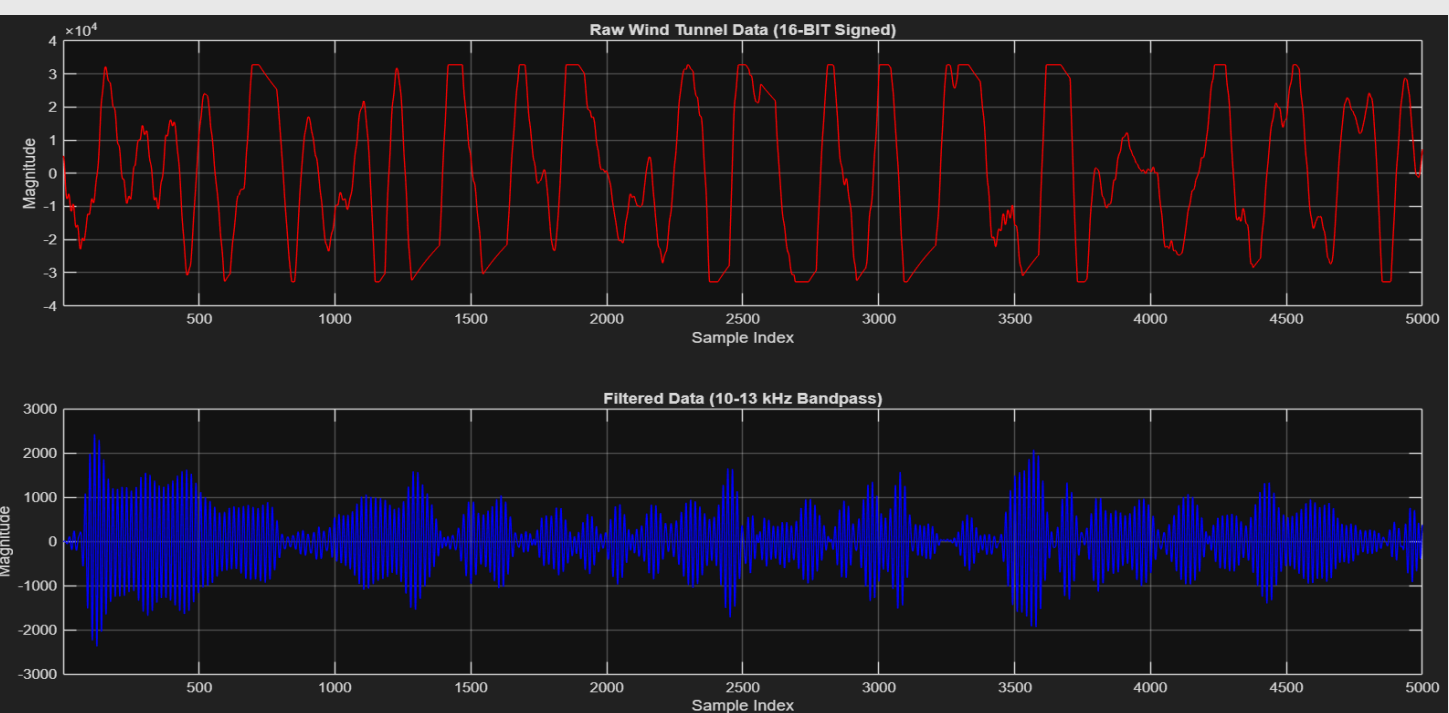


Figure 4. Time-Domain Plot of Raw Acoustic Data Collected by the Analog MEMS Microphone at the Aeronautical Wind Tunnel and the Filtered Data from the Standard FIR Filter Simulated in MATLAB.

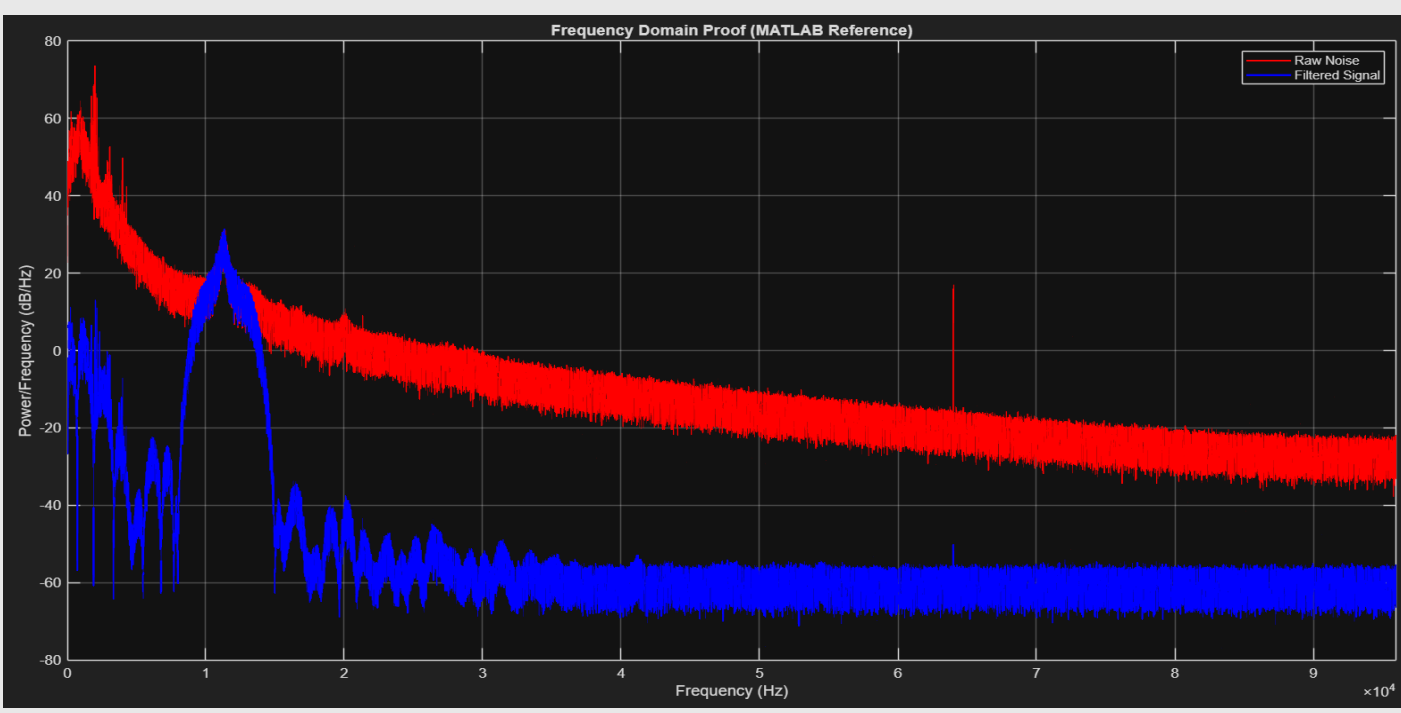


Figure 5. Power Spectral Density of Raw Acoustic Data Collected by the Analog MEMS Microphone at the Aeronautical Wind Tunnel Overlaid With the Simulated Filtered Signal From the Standard FIR Filter.

Optimized FIR Filter

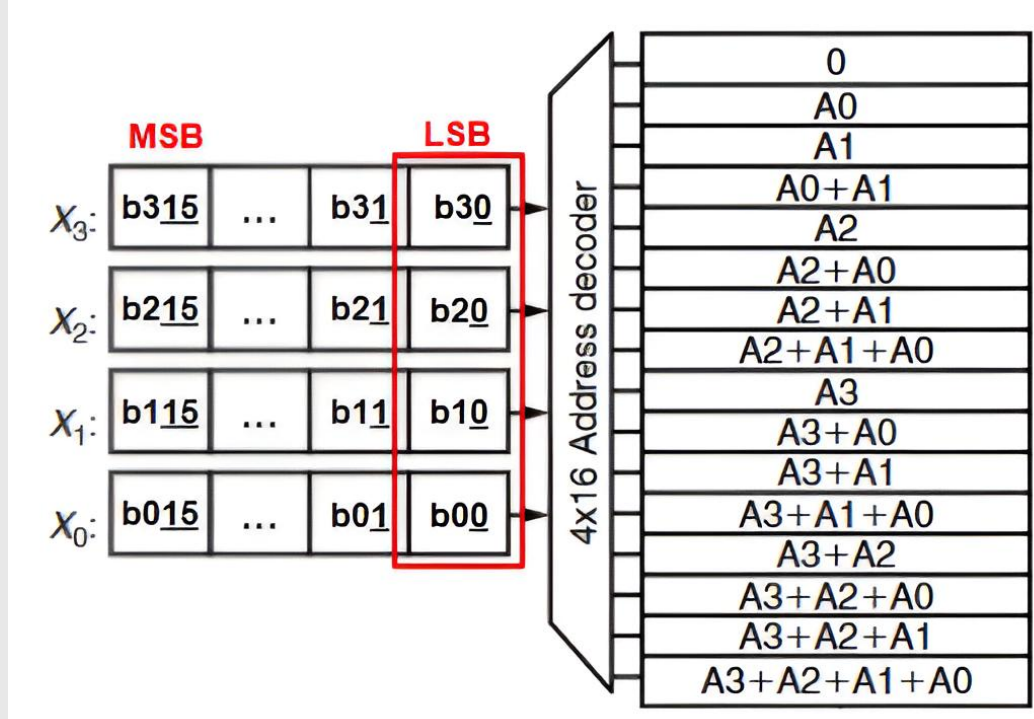


Figure 6. Block Diagram of Look-Up Table.

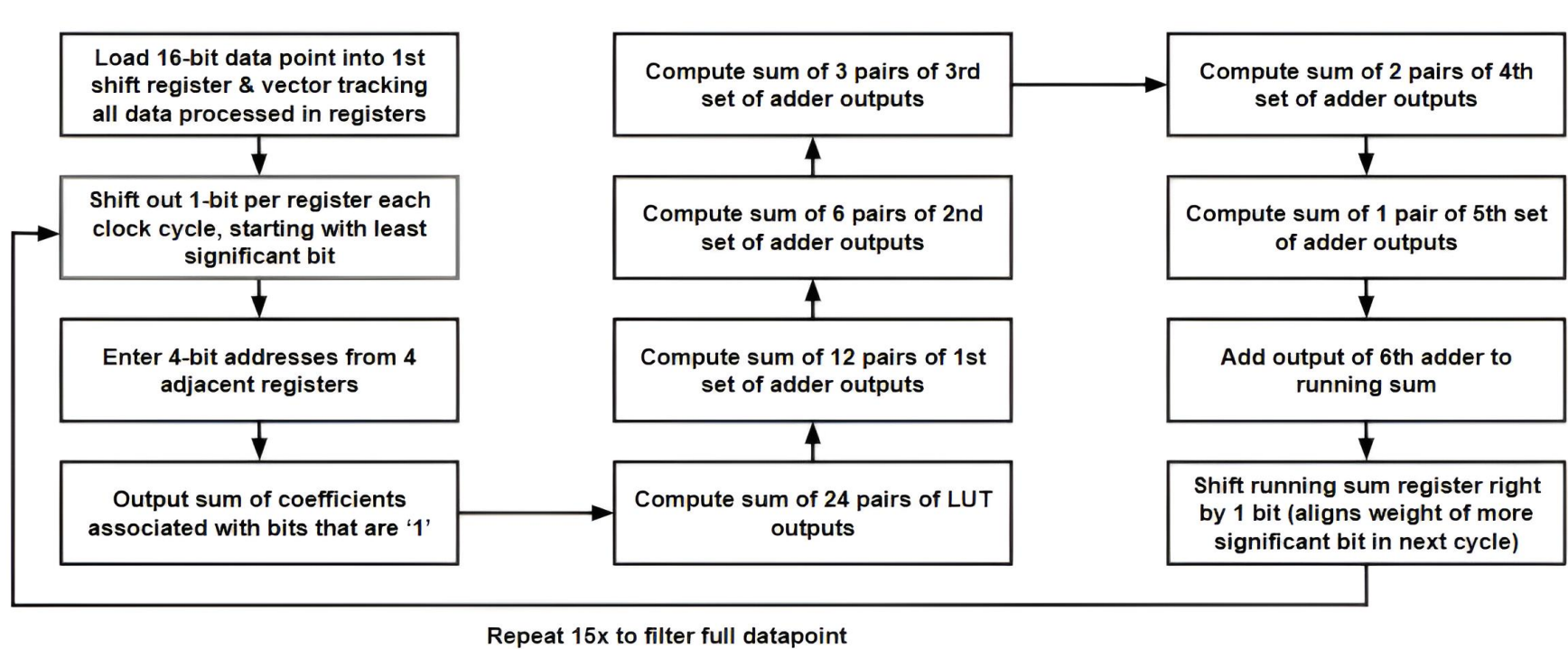


Figure 7. Flowchart of System Logic to Process One 16-Bit Data Point.

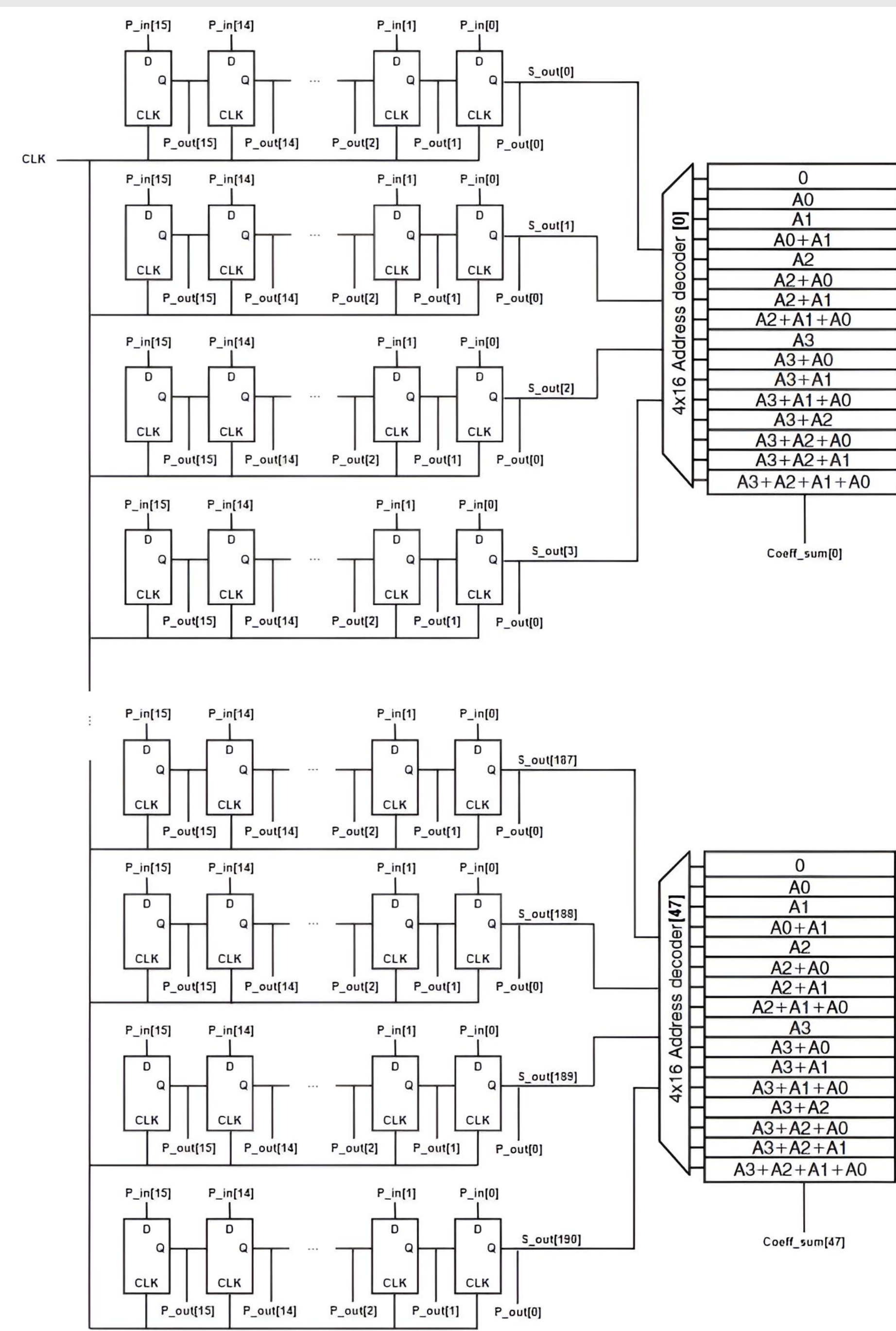


Figure 8. Block Diagram of Input Shift Array (191 16-Bit Parallel-In, Serial/Parallel-Out Shift Registers) and Address Search (48 4-Tap Look-Up Tables).

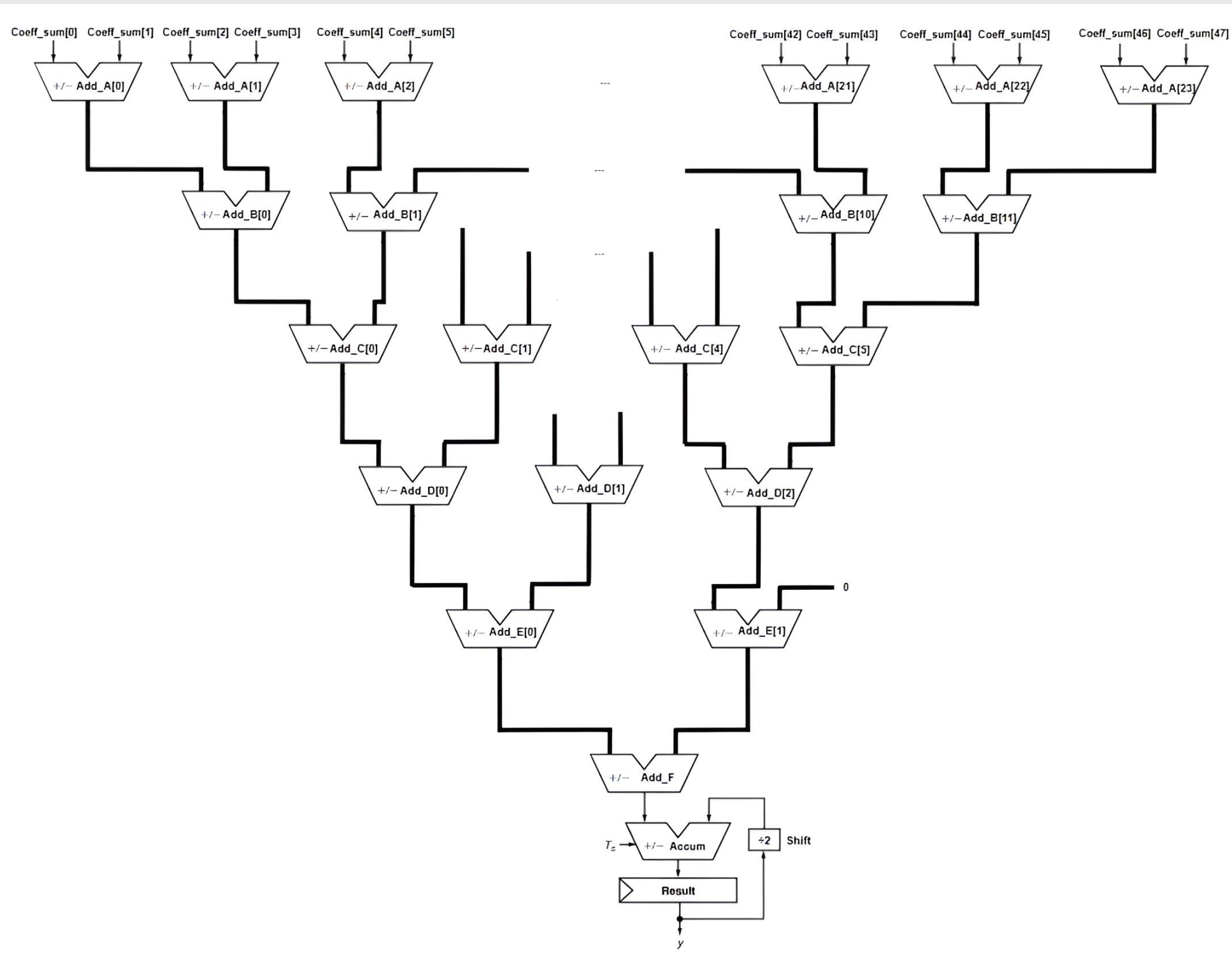


Figure 9. Block Diagram of Binary Adder Tree (48 2-Input Adders) and Accumulator.

Results

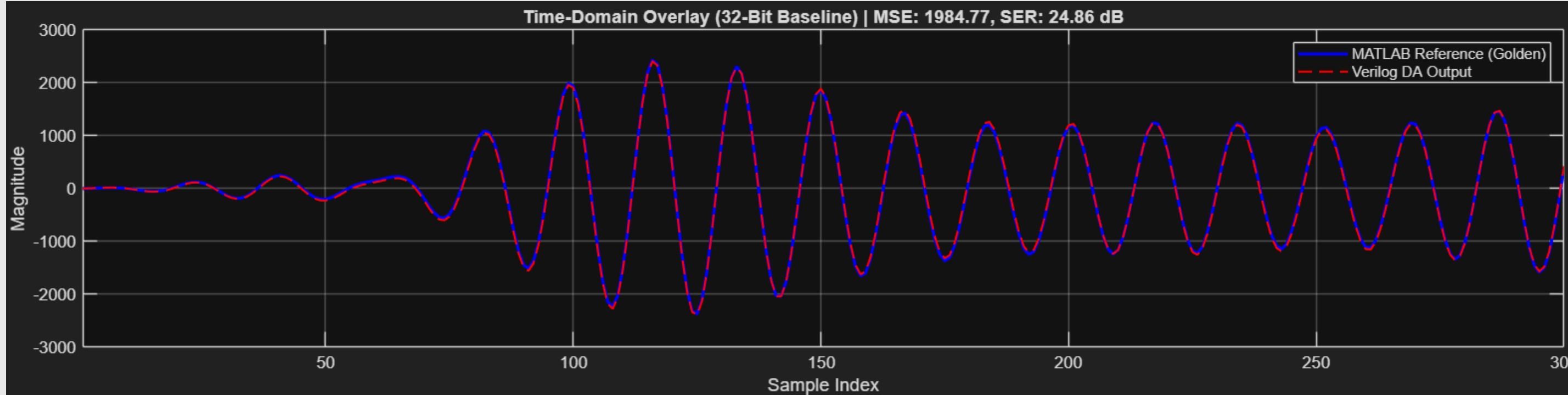


Figure 10. Time-Domain Plot of Filtered Acoustic Data Points From Standard and Optimized FIR Filters Simulated in MATLAB.

Power Dissipation Metric	MAC-Based Filter	DA-Based Filter	Power Reduction
Total Thermal Power	123.26 mW	105.69 mW	14.25%
Core Dynamic Thermal Power	11.36 mW	3.12 mW	72.54%
Core Static Thermal Power	90.05 mW	89.97 mW	0.09%
I/O Thermal Power	21.85 mW	12.60 mW	42.33%

Figure 11. Power Consumption Comparison Between Standard and Optimized FIR Filters.

Resource Metric	MAC-Based Filter	DA-Based Filter
Logic Cells	67	138
Dedicated Logic Registers	67	67
Memory Bits	2292	0
M9K Blocks	1	0
DSP 18x18 Blocks	1	0
LUT-Only Logic Cells	0	69
Register-Only Logic Cells	66	23
LUT/Register Logic Cells	1	46

Figure 12. Resource Comparison Between Standard and Optimized FIR Filters.

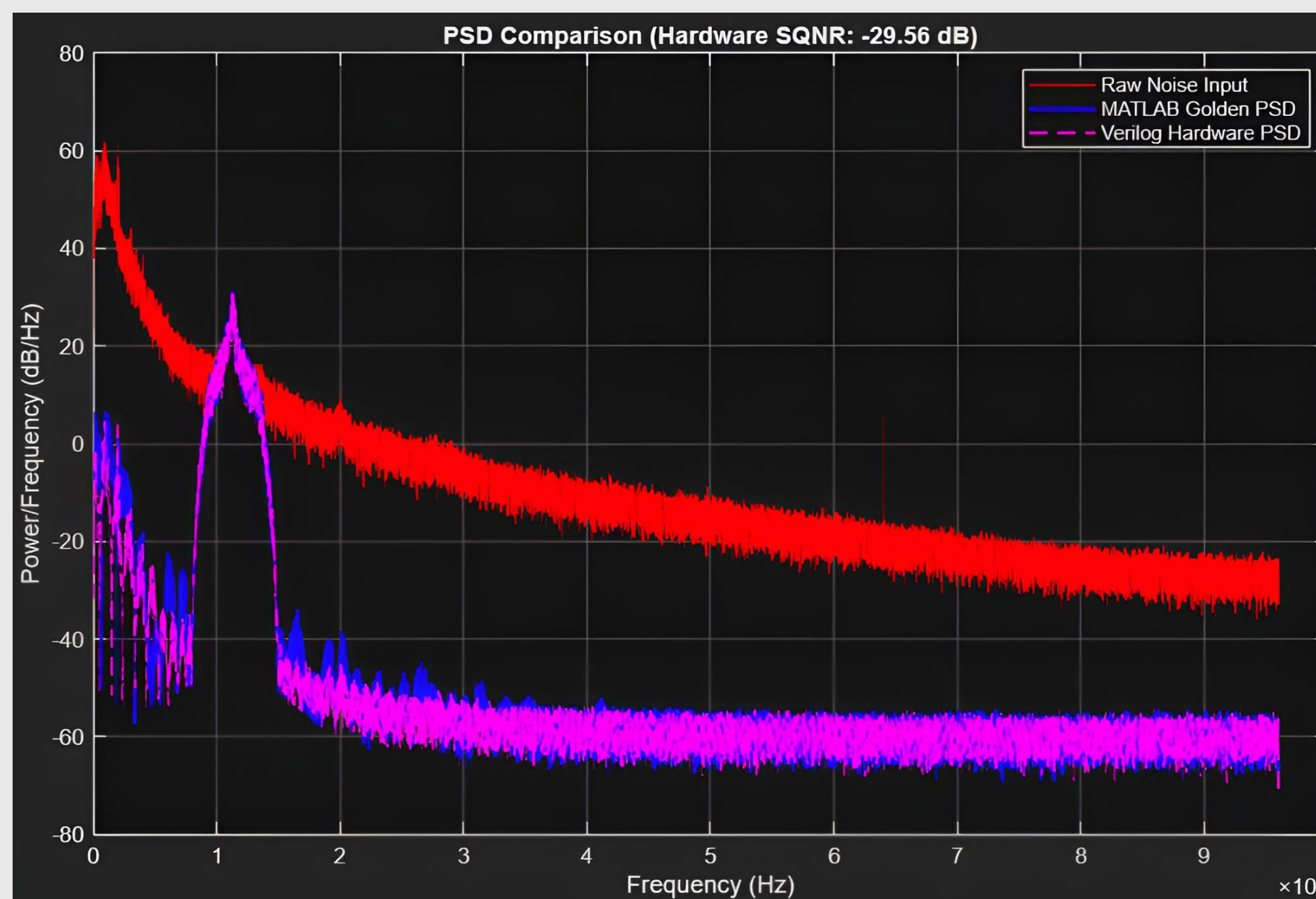


Figure 13. Power Spectral Densities of Raw Acoustic Data Overlaid With Filtered Data From Standard and Optimized FIR Filters.

Platforms



MATLAB



Quartus Prime



ModelSim

Acknowledgements

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References

- [1] STMicroelectronics, *Getting Started with the STEVAL-STWINBX1 SensorTile Wireless Industrial Node Development Kit*, User Manual UM2965, Rev. 7, Nov. 2024. [Online]. Available: https://www.st.com/resource/en/user_manual/um2965-getting-started-with-the-stevalstwinbx1-sensortile-wireless-industrial-node-development-kit-stmicroelectronics.pdf.
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